



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment 6

Vacuum Cleaner Problem

Aim

To write a Python program to implement the Vacuum Cleaner Problem using a simple reflex agent.

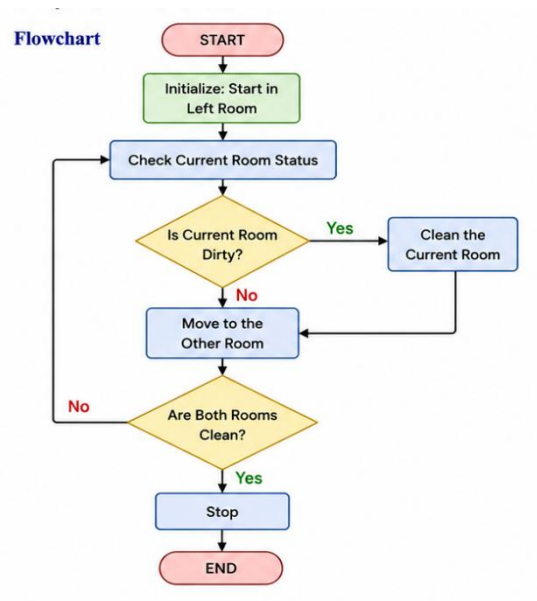
Objective

The objective of this experiment is to implement a simple reflex agent for the Vacuum Cleaner Problem. It aims to clean all dirty rooms by performing appropriate actions based on the current environment.

Algorithm

1. Initialize the vacuum cleaner in the left room.
2. Check whether the current room is clean or dirty.
3. If the room is dirty, clean it.
4. Move to the other room.
5. Repeat the process until both rooms are clean.
6. Stop when all rooms are clean.

Flowchart



Code

```
class VacuumCleaner:
```

```
    def __init__(self, rooms):
```

```
        self.rooms = rooms
```

```
        self.position = 0
```

```
    def clean(self):
```

```
        while True:
```

```
            if self.rooms[self.position] == "Dirty":
```

```
                print(f'Cleaning Room {chr(65+self.position)}')
```

```
                self.rooms[self.position] = "Clean"
```

```
            else:
```

```
                print(f'Room {chr(65+self.position)} is already clean')
```

```
            if all(room == "Clean" for room in self.rooms):
```

```
                break
```

```
            self.position = 1 - self.position
```

```
print("\nFinal Room Status:")

print(self.rooms)

rooms = ["Dirty", "Dirty"]

agent = VacuumCleaner(rooms)

agent.clean()
```

Output

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 6.py =====
Cleaning Room A
Cleaning Room B

Final Room Status:
['Clean', 'Clean']
```

Result

The Python program successfully implemented the Vacuum Cleaner agent and cleaned all the dirty rooms.

Conclusion

The Vacuum Cleaner Problem demonstrates the working of a simple reflex agent that senses the environment and performs actions to achieve the goal of cleaning all rooms.